

Defeating Cerberus: Privacy-Leakage Mitigation in Vision Language Models

Anonymous ACL submission

Abstract

Vision Language Models (VLMs) have demonstrated remarkable capabilities in processing multi-modal data, but their advanced abilities also raise significant privacy concerns, particularly regarding Personally Identifiable Information (PII) leakage. While relevant research has been conducted on single-modal language models to some extent, the vulnerabilities in the multi-modal setting have yet to be fully investigated. Our work assesses these emerging risks and introduces a concept-guided mitigation approach. By identifying and modifying the model’s internal states associated with PII-related content, our method guides VLMs to refuse PII-sensitive tasks effectively and efficiently, without requiring re-training or fine-tuning. We also address the current lack of multi-modal PII datasets by constructing various ones that simulate real-world scenarios. Experimental results demonstrate the method can achieve on average 93.3% refusal rate for various PII-related tasks with minimal impact on unrelated model performances. We further examine the mitigation’s performance under various conditions to show the adaptability of our proposed method.

1 Introduction

Large language models (LLMs) have demonstrated promising performance across multiple domains. Real-time AI assistance built with these models, such as ChatGPT¹ and Copilot², are already deployed for commercial use. The recent emergence of multi-modality in such models has further expanded their capabilities. Especially for scenarios that combine language and vision, which are two of the most common channels humans process information, LLMs have been utilized as the backbone to construct Vision Language Models (VLMs).

¹<https://chatgpt.com/>

²<https://github.com/features/copilot>

Traditionally, many approaches for multi-modal tasks use distinct and separate models for processing different modalities of data before combining each step into a comprehensive pipeline (Laina et al., 2019; Ngiam et al., 2011). In contrast, newer models can directly process different modalities of data within a single model or input pipeline (Zhu et al., 2023; Bai et al., 2023; Liu et al., 2023a). For example, instead of first converting an image into a textual description and then conducting downstream tasks based on that description, VLMs can directly process instructions that incorporate both text-based commands and target images. These new VLMs can outperform previous systems that rely on other types of models for a wide range of tasks (Bang et al., 2023; Yin et al., 2023).

However, these multi-modal capabilities can also be exploited for malicious purposes. For the backbone LLMs in these VLMs, there are already emerging attacks that specifically target the model’s ability to process instructions and understand complex context (Gu et al., 2024; Xie et al., 2023; Zou et al., 2023b). These attacks can “trick” these LLMs into performing policy-violating or harmful actions. In the privacy domain, Personally Identifiable Information (PII) has been a particular focus for the attacks targeting these multi-modal models. Given their strong generative abilities, these models may potentially reproduce privacy-violating materials that were used during their training or fine-tuning. Furthermore, even when leakage of private information from training data is not a concern, these advanced models can conduct (potentially harmful/illicit) PII-related tasks at scale.

The additional visual input in VLMs presents another surface that can be further exploited to expose these vulnerabilities. While these risks have been examined for LLMs (Huang et al., 2022; Lukas et al., 2023), similar vulnerabilities in newer multi-modal models are yet to be thoroughly investigated.

Compared to LLMs, investigating these risks

for VLMs poses several new challenges. First, although many native LLM/VLMs have existing safety guardrails that deter their utilization for harmful/policy-violating results, auxiliary attacks, such as jailbreaking (Zou et al., 2023b; Deng et al., 2023; Liu et al., 2023c) or backdoors (Huang et al., 2023; Xu et al., 2023; Yan et al., 2023), can successfully bypass these defense mechanisms. Worse, the vision modality of VLMs introduces additional channels for injecting malicious triggers for these attacks. Second, the visual input to a VLM can be highly variable, including, but not limited to, different shapes, concepts and objects. As a result, any mitigation mechanism needs to be highly adaptable and should not affect benign task performance. Finally, the evaluation of such mitigation mechanisms requires corresponding test datasets. Even though there are several datasets involving PII, these datasets are all in text format. In contrast, in the context of multi-modal models, the test datasets should also be in a multi-modal format (e.g., text *and* images for VLMs). Constructing such datasets realistically is not a trivial task.

To address these gaps, we investigate the potential risk of PII leakage in VLMs and propose corresponding mitigation methods. We first address the lack of test datasets by constructing realistic multi-modal versions of existing text PII datasets that simulate real-world use cases, such as document scans and ID cards.

We then draw inspiration from recent developments (Zou et al., 2023a; Arditi et al., 2024) in interpretable machine learning to develop suitable mitigation mechanisms for deterring PII leakage from VLMs. In our approach, we identify model weights that are mostly associated with PII. We then edit these weights, so that the models become more attentive to the *concepts* of generating PII-related content. The modified model now refuses to comply with requests that involve PII.

Our results show that we can effectively deter VLMs from executing tasks related to PII in various scenarios, reaching a refusal rate of 93.3% on average with minimal impact on unrelated tasks. The multi-modal dataset, as well as the mitigation mechanism, will be made open-source.

2 Background and Related Work

2.1 Vision Language Models

The generative capabilities of LLMs have been extended to other modalities with multi-modal mod-

els. These extended capabilities further allow the adoption of such models in various new scenarios and domains. Vision and language processing are two of the most prominent fields in the Machine Learning (ML) community, enabling the integration and development of multi-modal models for application to various domains. As such, Vision Language Models (VLMs) represent a prominent branch of the multi-modal LLMs.

Most of the VLMs to date (Liu et al., 2023a; Zhu et al., 2023; Liu et al., 2023b) leverage a single-modal LLM as their backbone and intend to incorporate the visual information directly as inputs to the backbones. The key component in these models differs primarily in how the image and its information are incorporated with the text command and input to the backbone LLM. Similar to the way the text inputs are encoded into embeddings before generating downstream responses in an LLM, the image input can also be encoded into corresponding embeddings that can be “understood” by the model.

2.2 Personally Identifiable Information

Defining Personally-Identifiable Information (PII) precisely can be challenging. According to the General Data Protection Regulation (GDPR), PII includes all types of information that are related to an identified or identifiable natural person. One potential challenge is that different contexts or scenarios can affect what is actually important in protecting the information owner’s privacy. Therefore, the design for corresponding leakage mitigation should also be flexible. We refrain from attempting to define precise PII since it is outside our scope. We simply select various types of potential private personal information (e.g., contact information, job), and conduct experiments on multiple types to further demonstrate our method’s versatility.

2.3 PII-leakage Risks of LLMs

Given the generative capabilities of LLMs, the leakage of PII that was inside the training datasets becomes a potential issue that can lead to vulnerabilities in exposing private information. For example, previous works (Huang et al., 2022; Lukas et al., 2023) have investigated such risks at different stages, such as pre-training and in-context learning.

Besides leaking sensitive private data that are used for training and fine-tuning, allowing LLMs to execute tasks involving PII can also introduce potential risks. Although many LLMs are only ca-

pable of utilizing the information in their training datasets, recent advances enable LLMs to also utilize external tools (e.g., web/database search) for giving more up-to-date and involved responses³. With the ability to utilize external tools and process complex commands, these models can potentially be used to extract PII from external sources. For example, an LLM can be prompted to search for specific private information referring to natural persons (Xi et al., 2023; Mo et al., 2024). These advanced capabilities allow them to easily outperform humans in scale when executing the same task (e.g., searching external sources), leading to a much bigger potential risk.

In light of these risks, many commercially available models have policies and guardrails against using them for PII-related tasks^{4 5}. In this work, we are particularly interested in investigating the potential of utilizing VLMs for PII extraction and mitigating their potential risks.

3 Multi-Modal PII Datasets

3.1 Existing PII Datasets

Before evaluating the potential risks of these models, we need to acquire realistic multi-modal PII data. While a sizable collection of PII datasets has been used in previous work, these datasets are all in text format, as expected. They can be separated into two categories: datasets generated from real-world data (e.g., Enron emails (Klimt and Yang, 2004)), and synthetic datasets (Holmes et al., 2024). We need PII data that is in visual format to simulate realistic use cases of these multi-modal models. Due to the lack of existing datasets, we construct the datasets ourselves. We will make these datasets and their construction tools available to the community.

3.2 Constructing Multi-Modal PII Datasets

To construct a multi-modal PII dataset, obtaining relevant data can be challenging. For our focus on PII leakage from VLMs, ideally, the datasets should consist of images of texts that contain sensitive information (PII). Unlike text-based PII datasets, obtaining original images of documents that contain PII can be difficult, especially at scale. As for generating synthetic data, while current advanced text-to-image models can generate a high variety of images impressively, generating images

that contain accurate text as instructed can still be challenging. Even some of the most advanced commercial models cannot generate images that are realistic enough compared to actual images with legible text, let alone PII (see Figure 7 for examples). Therefore, for now, directly generating synthetic datasets from text-to-image models is unfortunately not viable.

To overcome these challenges, we adopt an alternative strategy and leverage the existing text-based PII datasets by converting them into multi-modal versions. Specifically, we use two approaches: 1) direct conversion and 2) context injection.

Direct Conversion. For direct conversion, as the name suggests, we convert the text-based PII data directly into image format. This approach is applicable in various real-world scenarios, in which hard-copy documents have been converted into digitized versions by scanning them. This kind of digitization is a common occurrence for modernizing archival infrastructure for governments and newspapers (e.g., NYTimes⁶) to create an easily searchable and maintainable database of various documents. To represent a similar effort, we can convert the text of the email content from the Enron dataset (Klimt and Yang, 2004) into images that represent scanned and digitized documents. For previous text-based synthetic datasets, we can also format the sensitive texts into tables or other variations that can potentially be used to present such data. We construct the PII-Table dataset that contains images of generated tables from synthetic PII datasets⁷, with samples shown in Figure 1.

For direct conversion, these images are usually simulating documents that include text that might contain PII. For these conversions, then, the key is simulating the realistic artifacts created by the conversion tool (e.g., dust particles in scanned documents). We further improve the realism of such simulations by adding additional manipulations that simulate noises and artifacts introduced to the image when first converted from actual documents (e.g., scans and photos). We use the common open-source library OpenCV to generate these manipulations. For the direct conversion dataset we generated, we also constructed manipulated versions with different types and degrees of disturbance added, as shown in Figure 2.

Context Injection. While direct conversions can

³<https://openai.com/research/gpt-4>

⁴<https://www.anthropic.com/legal/aup/>

⁵<https://openai.com/policies/usage-policies/>

⁶<https://www.nytimes.com/>

⁷<https://huggingface.co/datasets/ai4privacy/>

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Figure 1: PII-Table Dataset Sample.

name	email	phone	job	address
Aaliyah Popova	aaliyah.popova4783@aol.edu	(95) 94215-7906	jeweler	97 Lincoln Street
Konstantin Becker	konstantin.becker@gmail.com	0475 4423797	developer	826 Webster Street
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Figure 2: PII-Table Dataset Sample with “Scanned” Effect.

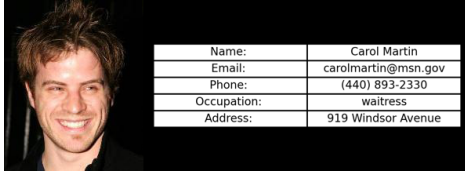


Figure 3: CelebA-Info Dataset Sample.

simulate potential documents involving PII texts, the variety of the data can be limited. Besides direct conversion, we also construct context-injected multi-modal datasets containing PII. Similar to generating synthetic datasets containing *only* text PII, we construct possible scenarios where *multi-modal data* (e.g., photos) might exist, such as scans of ID cards, professional resumes, and personal information tables. Utilizing additional open-source image datasets, such as CelebA dataset (Liu et al., 2015), we combine face images from the CelebA with randomly selected synthetic personal information, such as email, address, and phone numbers, to construct the *CelebA-Info* dataset, as shown in Figure 3. This type of context-injected data further expands the variability in multi-modal PII data, and combines the vision and language model in a more realistic and natural manner.

4 Internal Concept Steering

With LLMs becoming larger and increasingly sophisticated, previous works (Zou et al., 2023a; Ardit et al., 2024) have found comprehensible concepts based on the activities of models’ internal states. These concepts can range from tangible

entities, such as the Golden Gate Bridge⁸, to abstract notions, such as harmful behaviors (Zou et al., 2024) or refusal of requests (Arditi et al., 2024). By modifying the weights that are most active when these concepts are present, one can steer the model towards or away from them. Exploiting this observation, we suspect that we can modify the method to extract internal representations of PII and guide the models away from generating PII-related content.

Even though the models we study are multi-modal, the concept extraction and weight steering are conducted in a single modality (i.e., text). The vision component of the VLM is only responsible for processing the image input into embeddings that can be used as input to the backbone LLM. The backbone LLM is responsible for processing the information before generating the corresponding output. The concepts should exist within the LLM backbone regardless of the source of the input information (e.g., an address directly input through text to the VLM should provide the equivalent information to the backbone LLM as an image that contains that address).

4.1 Concept Extraction

The pipeline for extracting concepts from a model’s internal hidden states essentially involves drawing the model’s attention to the desired concept and observing the neuron patterns in the model. We first construct a demonstration dataset $\mathcal{D}_{demo}$ that includes positive samples $\mathbf{x}_i^+$ and negative samples

⁸<https://transformer-circuits.pub/2024/scaling-monosemanticity/>

$\mathbf{x}_j^-$, which correspond to sentences that include PII and ones that do not. To draw the model’s attention towards our desired concept, we use the following prompts before inputting the positive and negative samples respectively:

“Examine the following statement that contains sensitive/no private information:”

Notice that the defined “concept” encompasses more than just the entities of PII. It is a composite concept that recognizes these types of text as PII and acknowledges their sensitivity, where leakage could result in harm. This composite concept not only guides the model to identify PII but also activates internal guardrails to prevent potentially harmful content generation.

Instead of using generated results, we extract the model’s internal states $s_l(x_i)$ at each layer l for all samples in $\mathcal{D}_{demo}$ and obtain collections of internal states S for positive and negative inputs respectively:

$$S_l^+ = \{s_l(\mathbf{x}_i^+)\}, \quad S_l^- = \{s_l(\mathbf{x}_j^-)\}. \quad (1)$$

By randomly pairing positive and negative samples, we compute all the differences in their internal states to obtain set $\mathcal{D}_\Delta^l$ for each layer:

$$\mathcal{D}_\Delta^l = \{\Delta_{ij}^l = s_l^i - s_l^j \mid s_l^i \in S_l^+, s_l^j \in S_l^-\}. \quad (2)$$

We perform Principal Component Analysis (PCA) on the high-dimensional differences $\mathcal{D}_\Delta^l$ to find the principal direction $\mathbf{v}_l$ that maximizes the variance of all the collected differences:

$$\mathbf{v}_l = \underset{\|\mathbf{v}_l\|=1}{\operatorname{argmax}} \sum_{\Delta_{ij} \in \mathcal{D}_\Delta} (\mathbf{v}_l^\top \Delta_{ij})^2. \quad (3)$$

Ideally, the principal component $\mathbf{v}_l$ will represent the direction in the model’s internal state space at layer l that is aligned with the concept.

4.2 Model Steering

Given the directional vector $\mathbf{v}$, we can now *steer* the model towards or away from the concept. If we modify the model’s weights in the direction $\mathbf{v}$, the model should become less inclined to comply with requests that involve PII. By selecting a few layers that are the best at extracting the concepts (see subsection 5.2 for details), we modify the model weights through linear combination with the direction vector $\mathbf{v}$ and coefficient c :

$$\mathbf{W}_{new}^l = \mathbf{W}^l + c \cdot \mathbf{v}_l \quad (4)$$

Since we directly modified the model weights, the model with mitigation will not incur any additional computation cost at inference time.

5 Multi-Modal PII Leakage Mitigation

We now evaluate our proposed PII leakage mitigation method through concept extraction and model steering.

5.1 Experimental Setup

Models. For our experiments, we utilize Llava-Next (Liu et al., 2023a) as the VLM framework, which is a popular open-source architecture that has been widely examined in previous works (Liu et al., 2024; Gong et al., 2023; Gu et al., 2024). Within the Llava-Next framework, we evaluate several different backbone LLMs, including Mistral-7B (Jiang et al., 2023), Vicuna-7B, and Vicuna-13B (Chiang et al., 2023).

We also explored other VLM frameworks, such as MiniGPT-4 (Zhu et al., 2023) and Llava (Liu et al., 2023b). However, neither framework achieved acceptable performance on our target tasks. These VLMs struggle to effectively extract textual information from image inputs and exhibit significant issues with hallucination. For instance, when prompted with multiple *different* images from our CelebA-Info dataset, we observed that these VLMs output the *same* generic unrelated answers.

Datasets. We mainly focus on two of the datasets that we have constructed in section 3, namely PII-Table and CelebA-Info (with 1000 samples each). We also examine the versions with the “scanned” effect. For the demonstration set, we use an existing text-based PII dataset (Holmes et al., 2024), with 2000 samples for demonstration and 1000 samples for testing the concept extraction performance. These datasets contain PII of various types. We primarily focus on three targets that can be commonly considered PII: addresses, emails, and phone numbers.

Besides the multi-modal PII datasets, we also need unrelated (non-PII) tasks to evaluate the utility of the modified models. We use VHTest (Huang et al., 2024), a multi-modal dataset for evaluating VLMs’ hallucination behavior. The dataset includes a wide variety of open-ended questions for examining VLM’s capability of extracting information from the image inputs. For each run, we randomly select 500 samples for testing.

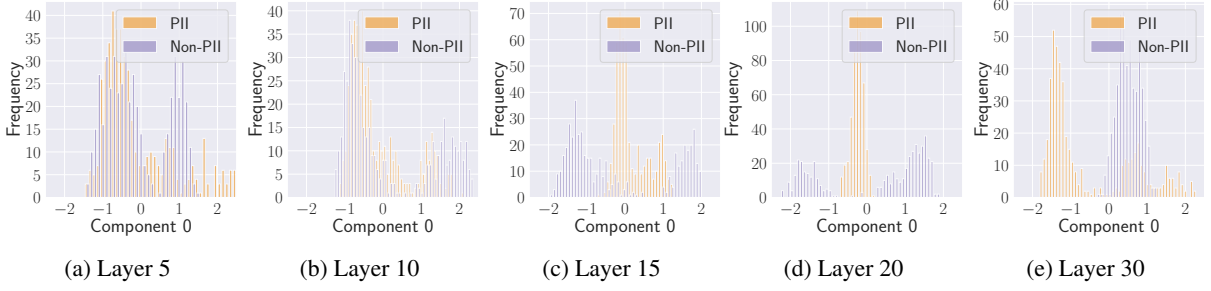


Figure 4: Distributions of test samples’ internal states’ projections on the principal component at different layers.

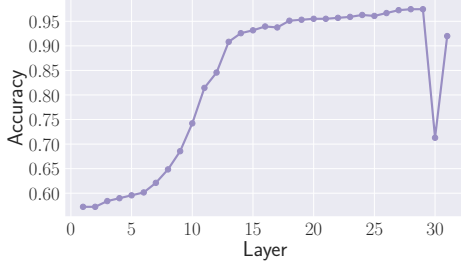


Figure 5: Concept extraction performance by internal states’ location (layer) on Vicuna-7B.

Metrics. To measure mitigation success rates, we construct a series of questions/tasks that aim to elicit PII from the image input. (For more detail see [Appendix B](#) in Appendix.) Since our focus is on leakage prevention, we refrain from evaluating these VLMs’ Optical Character Recognition (OCR) performance. Instead of inspecting whether the output contains the exact target PII, we confirm whether the model refuses to respond to the requests. A successful mitigation will prompt the model to refuse the user’s request, citing concerns about privacy violations and sensitive data leakage. We search for typical phrases used in such refusal responses to confirm mitigation effectiveness.

This method also allows us to directly evaluate the false positive rates on benign (i.e., non-PII-related) tasks. We also include nonsensical outputs from the model as “refusal.” (This can occur when the weights are modified too much.)

5.2 Concept Extraction Performance

We first examine the PII concept extraction performance from the models. This step serves two purposes. One is to confirm that the model has internal representations of our target concept. Two is to pinpoint the locations within the internal states of the model where they are most related to the concept so that we can effectively control the model’s behavior in the steering step.

To test the effectiveness, we collect internal states of an auxiliary dataset $\mathcal{D}_{aux}$ which is not part of the demonstration set $\mathcal{D}_{demo}$. Given these new internal states $\mathbf{h}$, we project them onto the direction vector $\mathbf{v}$ (at each layer) to compute the scalar value α which can represent the input sample’s correlation with the concept. We then calculate how accurately these projections can be used to predict whether the sample input is related to PII or not.

[Figure 5](#) shows the prediction accuracy based on the internal states after each layer. Overall, the prediction accuracy is very high, reaching over 95%. This implies that the model does have internal representations of PII. The prediction is especially accurate when using internal states from the later layers.

We can further visualize the effectiveness based on the distribution of projection values for all the test samples, as shown in [Figure 4](#). The projection values in the earlier layers (e.g., [Figure 4a](#), [Figure 4b](#)) do not show a distinct difference in the distribution between PII and Non-PII samples. However, in the later layers (e.g., [Figure 4d](#), [Figure 4e](#)), the two types of samples can be easily separated based on the projection values.

In addition, we also experiment with reducing the high-dimensional internal states’ differences to two principal components. Using the projection in 2D space we can better visualize how well the model can extract these concepts, as shown in [subsection D.2](#) (in Appendix).

As a result, we select the later layers as the targets for steering in the next step, specifically layers 15 to 25 for the Vicuna-7B backbone.

5.3 Model Steering Performance

While the projection values indicate that the models possess internal representations of PII (and related tasks), we now examine whether “steering” the model according to the directional vector can effectively limit its performance on PII-related tasks

	Mistral-7B			Vicuna-7B			Vicuna-13B		
	PII-Table	CelebA-Info	VHTest	PII-Table	CelebA-Info	VHTest	PII-Table	CelebA-Info	VHTest
No Defense	0.000	0.018	0.000	0.000	0.018	0.000	0.000	0.002	0.000
System Message	0.000	0.294	0.000	1.000	1.000	1.000	1.000	1.000	1.000
In Prompt	0.652	0.506	0.000	0.813	0.837	0.000	0.919	0.665	0.007
Ours	1.000	0.954	0.013	0.909	0.845	0.007	1.000	0.892	0.000

Table 1: VLM’s refusal rates on multiple tasks with various backbone models. PII-Table and CelebA-Info are datasets with PII (the higher the better for results). VHTest is non-PII datasets (the lower the better for results).

while preserving utility on unrelated tasks.

Baseline Comparison. As mentioned in section 2, we are not aware of any existing mitigation method that targets reducing PII generation from VLMs. Therefore, we include a comparison baseline stemming from a common defense strategy (Xie et al., 2023; Shen et al., 2024) deployed against other attacks against LLMs. This baseline defense injects a safety message either in the user prompt (*in prompt*) or within the *system message* of the model to “remind” the model not to execute PII-related tasks. These baseline defense methods are comparable to ours in setup since they do not require additional computing resources. For instance, using LLMs to judge the generated results could be another defense method (Phute et al., 2024; Zheng et al., 2023), but it requires additional inference. The resource requirement will also correlate with the generated output length.

From Table 1, we first observe that when no defense mechanism is deployed, the model will generally comply with users’ requests to generate PII-related outputs. For all models tested only less than 2% of such requests are refused. While the model does have guardrails for more malicious attacks, they are not tuned to refuse these requests.

Compared to the two types of baseline PII-Leakage mitigation methods, our method is the most effective on all datasets and backbone model types, without sacrificing utility tasks on benign tasks. For instance, our method achieves refusal rates of over 95% for both of the datasets on Mistral-7B backbone models, with only 1.3% of the unrelated tasks compromised. The best baseline defense can only achieve around 60% in the same setting. The baseline methods are more effective on the Vicuna family models. However, the mitigation is still not as effective as our method without significantly impacting normal model utility. For instance, when we inject the safety message into the Vicuna model’s system message, the model refuses to complete any request.

Model Variation. Table 1 also shows that the mitigation performance varies based on the backbone LLM. However, for all models examined, the mitigation is generally effective. On the lowest-performing backbone model, Vicuna-7B, and dataset combination, our method still achieves success mitigation on over 84.5% of the samples. Compared to the baseline methods, ours also has better consistency. The injected safety prompt’s effectiveness ranges from complete ineffectiveness to being too “effective,” where all tasks are refused. The model owner will likely need to carefully craft a safety prompt for each scenario and model setup. The lack of adaptability limits its practicality in real-world deployment.

Directly comparing performance on the same model architecture of different sizes, we can also see that the improved capabilities in larger models will also improve mitigation performance, as shown in Table 1 with Vicuna-7B vs. Vicuna-13B. The larger model has better concept extraction performance (see Appendix D in Appendix). Since we are only amplifying the model’s capabilities, we can expect a more powerful model to be better at concept extraction and subsequent steering.

Datasets. When comparing the two PII datasets tested, the mitigation performs well on both, though it shows a slight advantage on the PII-Table dataset, where the refusal rates are over 90% for all three models. Since the PII-Table dataset contains more concentrated PII, the model is understandably more sensitive to the presence of such private data. Further analysis of failed samples reveals that the image component in the CelebA-Info dataset can cause interference. The model occasionally prioritizes describing the person in the image and combines this description with the person’s name to make educated guesses about where they live. Although the model does not explicitly output the address from the image input, we still classify the mitigation as ineffective for more conservative results, as the model still complies with the request.

	PII-Table		CelebA-Info	
	Normal	Scanned	Normal	Scanned
Mistral-7B	1.000	1.000	0.954	0.941
Vicuna-7B	0.909	0.859	0.845	0.876
Vicuna-13B	1.000	0.998	0.892	0.875

Table 2: PII leakage mitigation performance on datasets with “scanned” effect.

	Address	Email	Phone
Mistral-7B	0.988	0.873	0.855
Vicuna-7B	0.997	0.350	0.733
Vicuna-13B	0.971	0.804	0.876

Table 3: PII leakage mitigation performance by types of PII.

When evaluating mitigation performance on samples with simulated “scanned” effects, the defense remains effective, as demonstrated in Figure 2. However, we observe that the introduced perturbations can impact OCR capabilities, sometimes leading to incorrect outputs.

The effective mitigation on multiple datasets and variations highlights the versatility of our methods. It is important to note that we do not adjust the steering settings between datasets. Once the appropriate layers and steering coefficients are set, the mitigation can be directly applied to any dataset.

Types of PII. We further conduct fine-grained analysis based on the type of the PII. Table 3 shows the refusal rates of concept-steered models on the CelebA-Info dataset based on the different types of target PII. The mitigation method is especially effective when the instruction aims to extract address information from the input images. The refusal rates are higher than 97% for all three models. The method, however, does not perform as excellent on email and phone number leakage mitigation. The performance is especially poor on mitigating email leakage from Vicuna-7B backbone model, with only 35% successful refusal. We suspect the model internally correlates personal addresses as more sensitive targets and thus such leakage is more easily mitigated. For the other two backbone models, the mitigation on these two types of PII is still generally effective, with over 80% refusal rates.

Steering Coefficient. Besides choosing the appropriate layers, it is essential to select the appropriate steering coefficient for optimal mitigation performance. When controlling the generation with the steering coefficient, we need to ensure sufficient

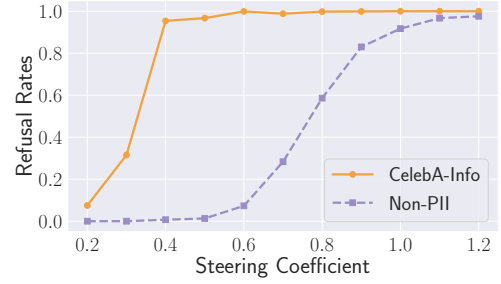


Figure 6: Steering coefficient affects mitigation and unrelated tasks’ performance.

mitigation magnitude while preserving the performance of unrelated (benign) tasks.

Figure 6 shows the modified Mistral-7B backbone model’s refusal rates of both extracting address information from the CelebA-Info dataset and executing non-PII tasks at different steering coefficients. The results show that the model’s refusal rates for both PII-related and benign tasks shift significantly within a narrow range of steering coefficients. Notably, there is a distinct gap between the coefficient values where mitigation performance declines and where disruptions to benign tasks become evident, at around 0.4 to 0.6. This behavior suits our mitigation application very well. It allows us to select the smallest coefficient right before the mitigation performance declines, minimizing the impact on normal task performance.

6 Conclusion

In this work, we address the critical need for understanding PII leakage in VLMs and effective mitigation strategies. Our concept steering approach demonstrates superior performance over existing methods on our constructed multi-modal PII datasets. As models continue to scale, the concept-steering mitigation offers both effectiveness and versatility without the need for training or fine-tuning. We hope our findings and datasets can further facilitate future research in this area.

Limitations

Our work is not without its limitations. First, the effectiveness of our mitigation method is contingent upon the capabilities of the target model. Specifically, our approach relies on extracting and amplifying existing “concepts” or “behaviors” already present within the model. If the model has not been fine-tuned with appropriate guardrails to refuse (any) potentially harmful requests, our mit-

igation strategy will be ineffective. Furthermore, models that are smaller or less capable may not have such internal representations of these concepts. However, we should expect future models (at least within the general transformer architecture frameworks) to be capable of constructing such concepts as they become more powerful.

Second, the mitigation pipeline requires manual adjustments to accommodate different models and application scenarios. While our method is versatile in targeting specific types of PII, the steering layer and coefficients need to be manually selected. This process involves iterating through various combinations during the initial development phase. However, since this process does not involve fine-tuning or retraining the model, the resource requirement is no different from deploying the model for inference usage. Ideally, the mitigation method can be improved with automated coefficient/layer selection. We defer such advancement to future work.

Ethics Statement

Given that our research concerns the critical and sensitive issue of personal, private information, we are deeply aware of the potential ethical implications. We conduct our analysis using publicly available data and models for both reproducibility and transparency. Additionally, to protect privacy, the PII data we used to construct our datasets and conduct experiments with are all synthetically generated and have open-source licenses. Recognizing the importance of this issue, we hope our proposed mitigation methods will further contribute to addressing these concerns.

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D Additional Concept Extraction Performance

D.1 Extraction Performance By Layers

Figure 8 showcases the PII-content prediction accuracy in test samples using projection values from Mistral-7B and Vicuna-13B backbone models. The prediction performance is especially high with the Vicuna-13B model, with over 80.0% accuracy in almost all layers.

D.2 Concept 2D Projections

The two-dimensional representation shown in Figure 9 generally agrees with results in Figure 4. However, for the ones inseparable in one dimension, we can still observe distinct, separable clusters in two dimensions, with each principal component representing the greatest variances in PII and non-PII data, respectively.



Figure 8: Concept extraction performance by internal states' location (layer).

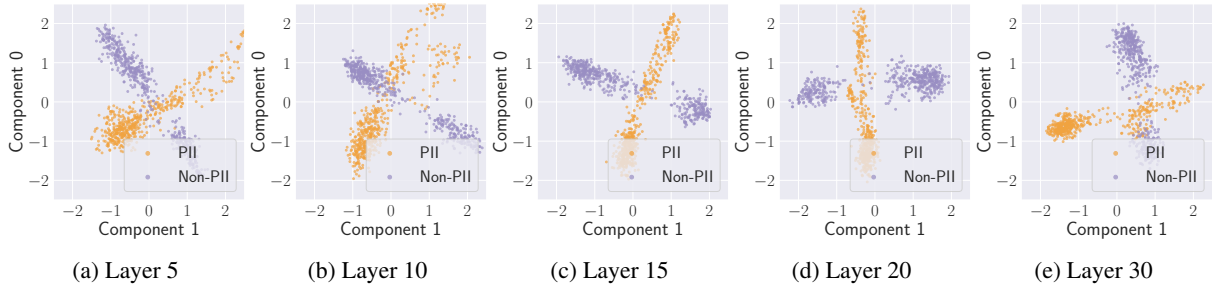


Figure 9: Test samples' internal states' projections on two principal components at different layers.